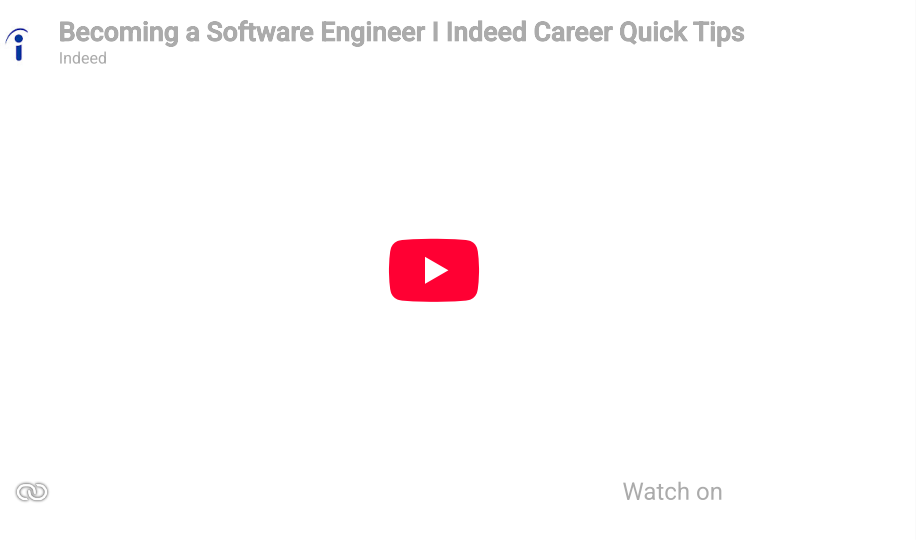
12 Software Developer Skills To Learn (With Examples)

Written by **Indeed Editorial Team**

Updated June 16, 2026



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Video: Becoming a Software Engineer

Are you interested in a career as a software engineer but are not sure how to start preparing for it? We have you covered!

KEY TAKEAWAYS:

- Many technical skills exist, including digital tools, programming languages, and computer systems, which software developers can learn through computer science degree programs or certification courses.
- By learning about different software developer skills, you can determine which option suits your preferences and can enhance your career opportunities.
- Software developers need to specialize in at least one type of coding language to increase the number of job opportunities they can receive, as hiring managers often expect candidates to have expertise.

Professional software developers have varied technical skills that allow them to succeed. They know how to operate multiple software programs, configure hardware devices and develop critical procedural steps to help them create their own applications. By learning about different software developer skills, you can determine which option suits your preferences and can enhance your career opportunities.

In this article, we discuss software developer skills and include 12 technical competencies you can gain.



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Source: Indeed Data, US, based on median, compared to job seekers who use mobile app

INSIGHTS FROM INDEED

Software Engineer Hiring Insights

Average pay

$135,994 / year

Low $32,000 **High** $312,000

Job seekers

230,246

+2,208 since August 2026

Employers hiring

4,121

+4,121 with open roles this month

Based on Indeed data · Refreshed monthly

See 13,327 open Software Engineer jobs →

What are software developer skills?

Software developer skills are the technical capabilities an individual learns to create successful programs and applications. They typically involve using mathematical knowledge and a capacity for problem-solving to write source code, which is the sequence of words and symbols that allows a program to function. Many technical skills exist, including digital tools, programming languages, and computer systems, which software developers can learn through computer science degree programs or certification courses.

Related: Software Engineer Skills: Definition, Examples and Tips

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Software developer skills

Here are 12 technical skills needed to become a software developer:

1. Coding languages

Coding languages are the different types of source code a person can write for a software program. Software developers need to specialize in at least one type of coding language to increase the number of job opportunities they can receive, as hiring managers often expect candidates to have expertise. To choose a language to learn, try to determine which type of program aligns with your interests. For example, some options may work better for mobile application development, while others may suit video game design instead.

Here are some common coding languages to learn:

- **JavaScript:** This option helps developers design the system for an interactive website application, such as a webpage that displays specific information after a user clicks a button.
- **HyperText markup language (HTML):** You can use this standardized option to display text and image content on a website application, such as paragraphs, bulleted lists, and charts.
- **Cascading styles sheets (CSS):** This option allows you to add fonts, colors and spacing to text content on a website application. As a result, many developers know HTML and CSS and use both languages simultaneously.
- **Python:** This option is a multi-faceted programming language that allows you to accomplish varied software development tasks, including writing code for an entire program or tracking errors in the language. Its simple structure may make it easier to learn than some options.
- **C++:** This programming language can help you develop programs that optimize hardware capabilities, such as medical equipment software. Many video game developers use C++ as they can design complex gaming systems with high-quality computer graphics.
- **Java:** This language can allow you to create programs and applications across different computer systems. Video game developers also use Java to create high-quality video games, as it has a similar structure to C++ with fewer complexities.

Related: 9 of the Most In-Demand Coding Languages (With Tips)

2. Database knowledge

Databases are electronic systems that allow individuals to organize and structure information, mainly company records. Software developers must know how to search for information, input new lines of data in a table format, and update a system with new materials.

They also understand how to secure a database and backup data to preserve a company's information during unforeseen circumstances. Learning a standardized source code called structured query language (SQL) may help operate a database, allowing you to operate varied systems for different companies.

Related: 10 SQL Skills For Programmers and Developers

3. Data structures and algorithms

Data structures are different organizational methods for storing information in a database, while algorithms describe the process required to complete tasks. Software developers often know many data structures and algorithms to determine which combination can help them optimize information in a program's code.

For instance, a completed array structure can store different variations of the same data type, such as a writing program that stores font information. If the developer uses the correct algorithm, users can choose a type of font and its size.

Learning different data structures may require a relevant course that teaches you how to build and implement them in a program. Here are some certification courses you can take:

- Coursera's Data Structures and Algorithms Specialization
- Udemy's JavaScript Algorithms and Data Structures Masterclass
- EdX's Professional Certificate in Data Structures and Algorithms
- Udacity's Data Structures and Algorithms Nanodegree Program

Related: **35 Common Data Structure and Algorithm Interview Questions**

4. Source control

Source control management (SCM) describes a system that helps a software development team track updates and changes to a program's coding language. It allows developers to work on the code separately and merge their efforts into one version. SCM systems can generate a list of code revisions and help you restore a language to a previous state if necessary. Learning how to use a type of SCM system may be helpful, as companies often require software developers to work in teams on the same coding projects.

Companies and individual developers often use Git, an SCM application that allows you to store different versions of program code in one folder system. It uses a branch system to organize each project, meaning a primary branch represents the original version of a code. In contrast, any additional branches that connect to it represent alterations or modifications. Git is also an open-source program, meaning it's available to anyone.

5. Testing procedures

Every software program requires an intricate testing process before a company can release it to consumers. The testing process uses digital tools to ensure a product functions and fulfills its design requirements. It also involves checking a program's ability to function on different hardware devices and store confidential information. As companies often design their testing processes based on a few critical components, it may be helpful for a software developer to understand fundamental aspects before applying to job positions.

Here are three components of software testing that developers know:

- **Unit testing:** This test involves evaluating individual lines or areas of code to determine if they require modification. Developers often create a different unit test for every project they complete to ensure they examine a program's key elements.
- **Integration testing:** Software developers use this process to determine whether they can successfully merge different areas of a program's code. They can also use integration testing to observe how one software program interacts with another's code.
- **System testing:** After integration testing is complete, an external testing professional typically conducts this process to determine if a product fulfills user expectations. A software developer needs to understand the steps of this process, as testing professionals often communicate critical suggestions for alterations.

Related: **Software Testing Methodologies: A Complete Guide**

6. Debugging knowledge

Debugging is a software development process for addressing and removing errors in a program's coding language. It involves running a program through specialized software to observe each error, then using problem-solving skills to determine which part of a code caused it. Many software developers debug their code while writing it to ensure it is functional during testing. Using debugging software is a fundamental strategy for removing errors and developing an effective process.

Related: **What Is the Debugging Process? (Plus 8 Strategies To Try)**

7. Operating systems

An operating system is software that configures a device's hardware and allows a program to run. Microsoft Windows, macOS, and Linux are the primary operating systems for computers, while iOS and Android are two standard options for mobile phones. It may be helpful for a software developer to develop expertise in the different types to conduct testing procedures better and transfer lines of code between them. Depending on your preferences and job position requirements, you can also specialize in developing program code for one system.

Related: Types of Operating Systems (With OS Functions and Examples)

8. Text editing software

Text editing software is a program that allows developers to create and edit plain text, which is a standard file that removes any formatting used in word processors. Software developers often use text editing software to write the text and symbols required for a coding language, as it allows them to transfer the code into another system and add formatting seamlessly. It is important to develop expertise in one type of text editing software so you can list it as a skill on a resume and streamline your software development process overall.

Here are some examples of standard text editing software:

- **Atom:** This option is open-source and optimized for software developers, meaning it provides access to Git and other useful program tools.
- **TextMate:** This software is compatible with macOS operating systems and includes interface support for all programming languages.
- **Notepad++:** This open-source software works on the Microsoft Windows operating system and has multiple customization options, including merging with another development program.
- **Brackets:** You can use this program on Microsoft Windows, macOS, and Linux operating systems and merge it with software that uses Git or similar development tools.

9. Cryptography

Cryptography obscures confidential data using an encryption code or password, often called a key. It involves using an algorithmic formula to develop a corresponding decryption code to reveal the original data to a user. Software developers usually use encryption techniques to prevent external agents from learning information about a product. Two primary types of cryptography to learn include symmetric encryption, which involves one key, and asymmetric encryption, which involves two keys to unlock the original data.

Related: How To Become a Cryptographer: A Complete Career Guide

10. Software frameworks

A software framework is an application developers can use to create a program using previously established information, such as lines of code you wrote or software knowledge from other developers. Each operating system has a different framework that typically incorporates the most common programming languages. They may include some predetermined structures and features that can streamline software code development. For example, you can use the .NET framework to create code on a Microsoft Windows system, Cocoa for a macOS computer system, or Cocoa Touch for an iOS mobile device.

Related: What Is a Framework? (Definition and Types of Frameworks)

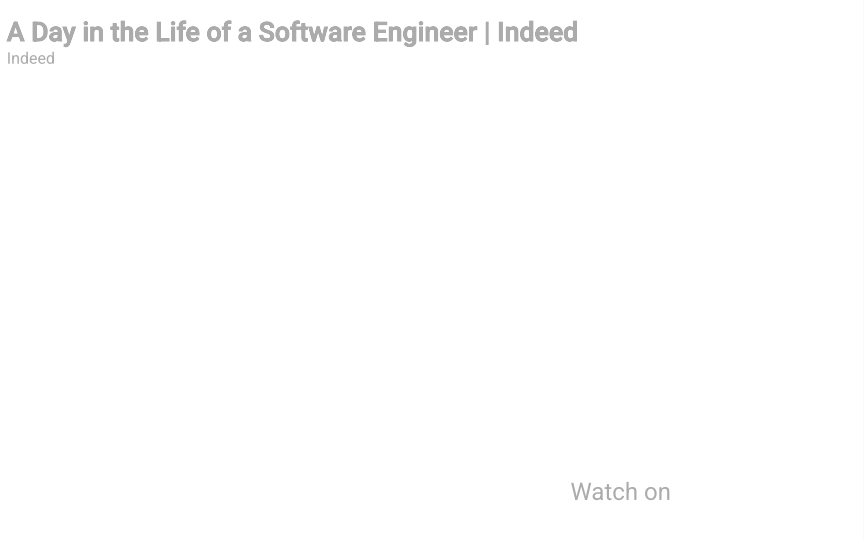
11. Cloud platforms

Cloud platforms describe the hardware and software required to run a company's remote database, now often called the cloud. Configuring this platform type involves knowledge about coding languages and server management, which are common skills for software developers. As many companies use these platforms to store information, it may be helpful to learn how to use programs and applications that use cloud-based technology. For example, you can become certified using Amazon Web Services (AWS), a cloud platform that multiple businesses purchase and use.

12. Spreadsheet software

Spreadsheet software allows individuals to organize data sets and make calculations. Learning how to use a spreadsheet program may be helpful, as a company may require software developers to track their data and project progress. You can also use a spreadsheet to solve any necessary equations or organize information before transferring it to a more extensive database system.

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Video: A Day in the Life of a Software Engineer

Sarah, a software engineer, shows you a day in the life of a remote software engineer including work hours, work environment and job duties.

Top 5 highly rated software companies to work for

(Based on employee reviews on Indeed)

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   Q: What do employees say about working at ADP?
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2. **TEKsystems**
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